

FINAL YEAR PROJECT PROPOSAL



ScholarsMate: An AI Assistant for Academic Synthesis

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Abstract

Keeping up with the mountain of new research papers is becoming an impossible task. Reading hundreds of pages of articles can often be time consuming and unproductive. While modern LLMs are great for chatting, they have a nasty habit of hallucinating i.e. producing incorrect or outdated information without any evidence to back it and they don't actually know what's inside your specific research papers unless they are trained on them. To fix this, we propose a Retrieval Augmented Generation (RAG) system designed specifically for academics. RAG connects LLMs to external, trusted data sources, preventing hallucinations by forcing the model to base answers on specific, provided content rather than internal training data. Instead of the AI guessing an answer or giving an outdated and unreferenced answer, it acts more like a librarian. It scans your uploaded papers, pulls out the exact relevant sections, and writes a response based only on those facts. It even provides citations which makes it easy to verify the answers. It's designed to speed up literature reviews and find research gaps. Overall it reduces the time consuming tasks by scanning, summarising and comparing papers for you without sacrificing academic integrity and irresponsible behaviour in academic work.

Contents

1	Introduction	3
2	Literature Review	3
2.1	Large Language Models and the Hallucination Challenge	3
2.2	Use of Retrieval Augmented Generation (RAG)	4
2.3	Academic Document Parsing and Embedding Techniques	4
2.4	Other Research Assistants	4
2.5	Summary of Research Gaps	4
3	Problem Statement	4
4	Objectives of the Project	5
5	Scope of the Project	5
5.1	In-Scope	5
5.2	Out-of-Scope	6
6	Proposed Methodology	6
6.1	Document Ingestion and Preprocessing	6
6.2	Vectorization and Knowledge Storage	7
6.3	Retrieval and Generation	7
6.4	System Architecture	7
7	Evaluation Criteria / Metrics	7
8	Tools & Technologies	8
9	Expected Outcomes	8
10	Project Timeline	9
11	Ethical and Security Considerations	10
12	Conclusion	10

1 Introduction

Every year millions of papers are published and just trying to stay current on new methodologies or theoretical debates has become a hurdle that researchers have to clear before they can even start their own original work. The traditional way of doing things manually such as reading through stacks of PDFs is not just too slow and prone to human error.

Large Language Models seem like the perfect fix for this. However, they are a general purpose AI and using them academic work is risky because of hallucinations where the model confidently makes up facts or citations that sound real but don't actually exist [2]. To fix this, Retrieval Augmented Generation has been introduced. This framework limits the AI's output to a specific, trusted set of documents. Unlike standard models that guess based on their internal training, a RAG system pulls relevant snippets directly from a (lets say) library and uses that context to build an answer. This approach keeps the AI's knowledge bounded by the provided papers, which drastically cuts down on hallucinations and allows for precise, page-level citations for every single claim.

But even with RAG, most tools still struggle with the messy reality of academic papers with stuff like multicolumn layouts, tables, and complex math. Most are also built for talking to one document at a time, which doesn't help when you need to synthesize an entire field or find research gaps across multiple papers.

This project addresses these gaps by creating ScholarsMate, a RAG-based assistant built specifically for academic work. ScholarsMate provides a workspace where researchers can interrogate their entire personal library at once. With features like automated literature, overview matrices and evidence mapping, the goal is to make scientific discovery faster without ever compromising on factual accuracy.

2 Literature Review

ScholarsMate is developed from the concepts Natural Language Processing (NLP), Information Retrieval, and Academic Knowledge Management. Here we review the evolution of Large Language Models, the role of Retrieval Augmented Generation as a solution to inconsistency, and existing academic assistant platforms.

2.1 Large Language Models and the Hallucination Challenge

Initial research into LLMs focused on increasing parameter scales and training data volume. However, as these models were applied to academic domains, people identified significant risks regarding factual integrity.

Table 1: LLM Limitations and RAG

Reference	Focus	Key Identified Gap
Huang, L. (2025) [2]	Hallucinations	Explained the hallucination types in LLMs.
Sriramanan(2024) [4]	Information Processing	Reduction of hallucination by chunking and evidence mapping.
Shuster et al. (2021) [3]	Knowledge Access	Proved that grounding reduces hallucination rates by over 50%.

2.2 Use of Retrieval Augmented Generation (RAG)

RAG was introduced to anchor generative outputs in verifiable external evidence. By retrieving relevant document snippets before generating a response, these systems change the nature of regular LLMs by providing the necessary context.

2.3 Academic Document Parsing and Embedding Techniques

Extracting knowledge from PDFs is a challenge for research assistants. Literature in this area emphasizes the difficulty of maintaining document structure (tables, headers, and citations) during the vectorization process.

2.4 Other Research Assistants

Platforms such as Elicit, SciSpace, and Consensus represent the current standard in AI-driven research. However each have their limitations [1]. A slight overview is give below:

Table 2: Comparison of Existing Academic AI Tools

Feature	Elicit	SciSpace	Consensus	NotebookLM
Primary Strength	Systematic Data Extraction	Literature Review Assistance	Evidence Synthesis (Yes/No)	Multi-source Synthesis
Search Engine	125M+ Papers	200M+ Papers	200M+ Papers	N/A (User-uploaded only)
Key Feature	Customizable Data Tables	Real-time Formula Explanation	"Consensus Meter" for evidence	Audio Overviews & Mind Maps
Analysis Focus	Methodology & Population	Deep-dive into single PDFs	Finding gaps in evidence	Long Context Understanding
Writing Output	Summaries & Abstracts	Paraphrasing & Snippets	Cited Statements	Study Guides & Notes

2.5 Summary of Research Gaps

Based on the review of current literature and existing tools, the following gaps have been identified:

- **Verifiability Gap:** Most general LLMs cannot provide clickable, page-level evidence for their claims, leading to user distrust.
- **Structure Awareness:** Some tools either have generic writing which cannot be tailored to niche fields while some required expert’s interpretation.
- **Synthesis Gap:** There is a lack of localized tools that can perform Literature Matrix comparisons across a user-defined folder of PDFs.

3 Problem Statement

Million of published papers have effected literature reviews. Researchers have to spend weeks manually sifting through thousands of papers before they can even start their own work. While AI tools like ChatGPT could help, they often hallucinate confidently inventing evidence because they rely on general training data rather than a specific set of papers.

To fix this, We propose a reliable assistant using Retrieval-Augmented Generation (RAG). Instead of the AI guessing, it pulls snippets directly from a researcher’s own PDF library to build its answers. This ensures every claim is factual in actual text and includes precise, page-level citations. Most

existing tools struggle with complex academic PDFs like multicolumn layouts and dense tables and usually only let you talk to one document at a time. The project, ScholarsMate, handles these technical hurdles and allows for multipaper synthesis.

Core Problem Statement: Eliminating AI hallucinations by using a framework that forces the model to pull answers directly from your own documents. It turns messy academic papers into a verifiable workspace where every insight is source based in real text, not AI guessing for you.

4 Objectives of the Project

The primary goal of this project is to develop ScholarsMate, a research assistant that utilizes Retrieval Augmented Generation (RAG) to provide verifiable, non-hallucinating insights from academic documents. The following objectives guide the development and evaluation of the system:

1. **Develop a Document Ingestion Pipeline** Extracting content from documents, not just text but also useful information like tables and equations.
2. **Establish a Semantic Knowledge Base** Using embeddings to form vectors, storing them in a database and matching vectors when needed
3. **Design a Factual Generation Architecture** Implement a system that restricts the LLM to generate responses solely based on retrieved document chunks, thereby eliminating generalized hallucinations.
4. **Implement Automated Literature Synthesis Features** Develop a comparative Literature Matrix feature that allows users to synthesize findings, methodologies, and results across multiple documents simultaneously. These are useful in academic work.
5. **Build an Interactive Evidence-Mapping Interface** Create a web based workspace that integrates all these features.
6. **Evaluate Retrieval and Factuality Performance** Assess the system using metrics such as retrieval precision, citation accuracy, and hallucination rates to ensure the reliability of synthesized information.
7. **Validate Academic Utility** Test the system on various research domains to ensure its effectiveness in identifying research gaps and accelerating the literature review process.

5 Scope of the Project

The scope of this project defines the technical and functional boundaries of ScholarsMate. The project focuses on creating a reliable environment for academic document interrogation and synthesis.

5.1 In-Scope

The following features and tasks are included within the scope of this project:

- **Document Processing:** Support for uploading entire folders of research papers in PDF format.
- **Semantic Search QnA:** A chat interface capable of answering specific questions based on the uploaded corpus.

- **Verifiable Citations:** Generation of citations linked to specific pages and text snippets in the source PDF.
- **Comparative Analysis:** Automated generation of literature review tables comparing different papers.
- **Local Data Persistence:** Implementation of databases for chat history and vector storage to ensure user data remains local and secure.
- **Table and Equation Parsing:** Basic extraction and interpretation of academic tables and mathematical formulas.

5.2 Out-of-Scope

The following items are explicitly excluded from this project:

- **Real-Time Web Searching:** The system will not use live internet search to avoid the introduction of unverified or non-academic information.
- **Automated Full-Paper Drafting:** The project provides research assistance but does not aim to generate entire research papers or write for the user.
- **Non-PDF File Formats:** Support is limited to PDF files, as they are the standard for academic publishing.
- **Fine-Tuning of Base LLMs:** The project utilizes existing models through RAG rather than performing resource intensive training or fine-tuning of large-scale models.
- **Direct API Integration with Paid Journals:** Users must provide their own PDF files; the system will not bypass paywalls or integrate directly with journal APIs.

6 Proposed Methodology

The proposed system follows a pipeline designed to ensure that the generative model operates exclusively within the bounds of the provided research documents.

6.1 Document Ingestion and Preprocessing

The first phase involves the extraction of data from a userprovided directory. This stage is critical for maintaining the structural integrity of academic content.

- **Recursive Folder Scanning:** Utilizing Python’s `os` or `pathlib` modules to identify and load all PDF files within a designated directory.
- **Layout-Aware Extraction:** Implementing **PyMuPDF (fitz)** to extract text while preserving structures such as headings and page numbers.
- **Recursive Character Chunking:** Text is divided into overlapping segments (e.g., 1000 characters with 100-character overlap) to ensure context is preserved across chunk boundaries for the embedding model.

6.2 Vectorization and Knowledge Storage

To enable semantic searching, the processed chunks must be converted into a machine readable format.

- **Embedding Generation:** Chunks are processed through a pre-trained transformer model (e.g., `text-embedding-3-small`) to create high-dimensional vector representations.
- **Vector Database :** These vectors, along with their associated text and metadata, are stored in a database like ChromaDB. This local database allows for rapid similarity searches.

6.3 Retrieval and Generation

The Chat interface utilizes a retrieval mechanism to prevent hallucinations.

- **Semantic Querying:** When a user asks a question, the system converts the query into a vector and performs a similarity search against the vector database to retrieve the top most relevant chunks.
- **Source-Locked Prompting:** The retrieved chunks are injected into another prompt that instructs the LLM to answer using only the provided text.
- **Citation Synthesis:** The model is instructed to append page and source references to every factual claim, derived from the data associated with the retrieved chunks.

6.4 System Architecture

The overall architecture of ScholarsMate follows a modular Retrieval-Augmented Generation (RAG) design, as illustrated in Figure 1..

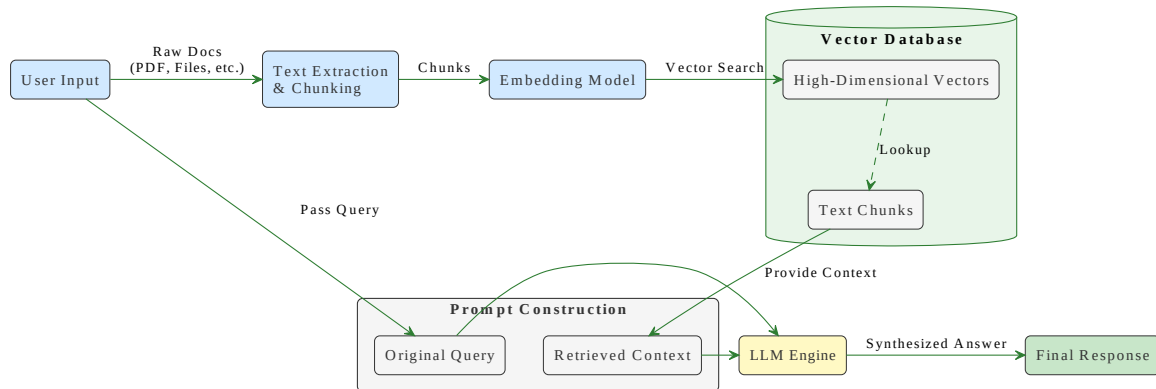


Figure 1: ScholarsMate System Architecture.

7 Evaluation Criteria / Metrics

The performance of ScholarsMate will be evaluated using metrics that measure both the accuracy of the retrieval process and the faithfulness of the generated answers.

- **Retrieval Precision@k:** Measures the proportion of the top- k retrieved chunks that are actually relevant to the user's query.

- **Faithfulness (Hallucination Rate):** Evaluates whether the LLM’s response is derived solely from the retrieved context. This is measured by comparing the claims in the answer against the source chunks.
- **Citation Accuracy:** Measures the percentage of generated citations that correctly point to the page and document containing the supporting evidence.
- **Answer Relevance:** A metric (often calculated via ”LLM-as-a-judge”) that assesses how well the generated response addresses the user’s specific research question.
- **Contextual Recall:** Measures the system’s ability to find all relevant information within the PDF folder necessary to answer a complex query.

8 Tools & Technologies

The following technologies have been selected to ensure a good research assistant experience.

Table 3: Tools and Technologies for ScholarsMate

Component	Technology	Description / Purpose
Frontend	React.js / Next.js	For building a responsive, interactive chat and document viewing interface.
Backend API	FastAPI	A high-performance Python-based API to handle PDF processing and LLM orchestration.
PDF Processing	PyMuPDF (fitz)	High-speed library for extracting structured text and metadata from research papers.
Vector DB	ChromaDB	A local-first database for storing document embeddings and enabling semantic search.
Relational DB	SQLite	Used for persistent storage of user chat history and project session data.
LLM Engine	Gemini / GPT-4o-mini/ Local Models	High-speed, context-rich models used for generation.
Embeddings	OpenAI / HuggingFace	Models used to convert extracted text chunks into high-dimensional vectors.

9 Expected Outcomes

Upon successful completion of this project, the following outcomes are expected:

1. **Academic Research Assistant** A functional web based system capable of providing non hallucinating answers derived strictly from user uploaded docs.

2. Structured Research Knowledge Base

A persistent local database containing high-fidelity text extractions, metadata, and high-dimensional embeddings for efficient multi-document retrieval.

3. Citation Framework

An integrated interface that maps claims directly to specific pages and highlighted snippets within the source PDF files.

4. Automated Synthesis Capabilities

A functional Literature Matrix feature that generates comparative tables of methodologies, results, and findings across multiple research papers.

5. RAG Performance Reports

Comprehensive evaluation data measuring retrieval precision, citation accuracy, and the reduction of hallucination rates compared to standalone LLMs.

6. Documentation and Source Code

Complete project documentation, including the RAG pipeline architecture and the full code for reproducibility and further research.

Additional Deliverables:

- A comparative study of different embedding models and chunking strategies to determine the most effective approach for academic document retrieval.
- A framework for identifying research gaps through automated cross-paper limitation analysis.

10 Project Timeline

The development of ScholarsMate is planned over a period of 6 months. This timeline accounts for the technical implementation of the RAG pipeline and final system refinement.

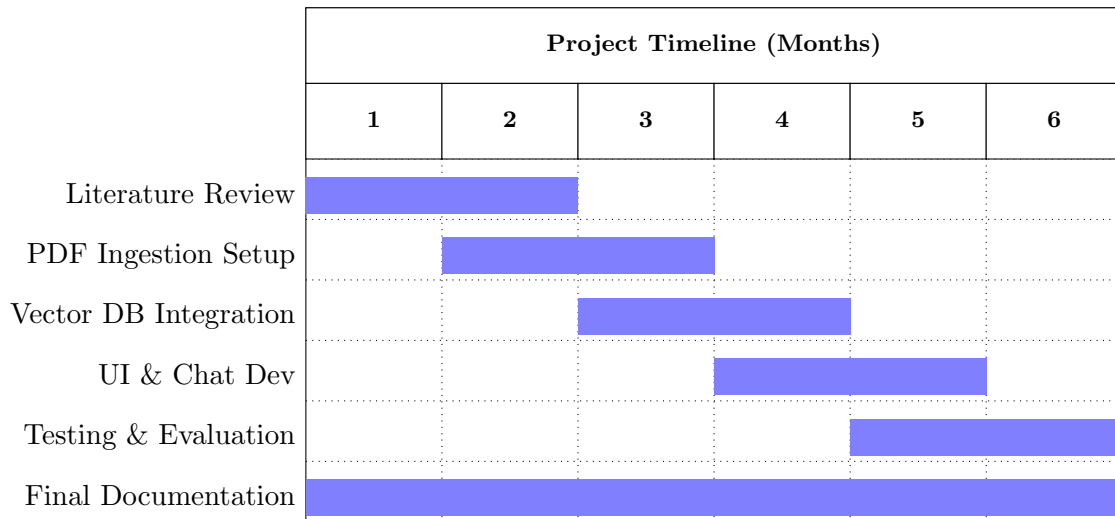


Figure 2: Project development timeline for ScholarsMate

11 Ethical and Security Considerations

- 1. Academic Integrity and Fact-Checking** The system is designed with a "source-locked" architecture specifically to mitigate the risk of AI hallucinations. By mandating direct evidence for every claim, ScholarsMate promotes academic honesty and reduces the spread of fabricated research data.
- 2. Data Privacy and Local Persistence**
To protect sensitive or unpublished research, the system utilizes local storage solutions (**SQLite** and **ChromaDB**). This ensures that the user's uploaded library and chat history remain within the local environment and are not used for retraining commercial LLMs.
- 3. Copyright and Intellectual Property**
ScholarsMate is a tool for synthesis and analysis it does not aim to bypass journal paywalls or violate copyright. Proper attribution and citation mechanisms are built into the model's output to ensure that original authors receive credit for their work.
- 4. Secure Code Management**
Access to the project repository and proprietary configurations will be restricted to authorized team members and supervisors.
- 5. Responsible AI Use**
The project is developed strictly for academic purposes. It provides assistance but does not generate full papers, ensuring that the critical thinking and final writing remain the responsibility of the human researcher.

12 Conclusion

ScholarsMate replaces generalized responses with a RAG architecture that anchors every response to your specific PDF library. By combining FastAPI, ChromaDB, and React, it creates a verifiable workspace where researchers can automate literature mapping without the risk of hallucinations or fake citations.

References

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